Smart-Factory Cloud Data Platform: Technical Summary

Tirthanker Singh · 28 October 2026 · Repository: https://github.com/TirthankerSingh/industrial-data-platform · Live dashboard: https://industrial-data-platform.streamlit.app · Full report: report.pdf

Scenario. An automotive Tier-1 supplier runs **100 CNC machines in three plants**: - **Chakan (Pune):** powertrain, 40 machines; - **Oragadam (Chennai):** transmission, 35 machines; - **Manesar (Gurugram):** chassis, 25 machines.

Each machine produces sensor, production and machine-control data, on Fanuc and Siemens controllers. The aim is to design and justify a cloud-data platform for monitoring, analytics, storage, dashboards, AI/ML and remote access.

Evidence base. Every number in this summary comes from **four public datasets** that were downloaded and processed: - **Bosch CNC vibration** (REAL): 1,702 operations sampled at 2 kHz; - **UMich CNC mill** (REAL); - **Microsoft Azure Predictive Maintenance** (BENCHMARK): 100 machines × 1 year; - **AI4I 2020** (BENCHMARK).

Values calculated from these are labelled DERIVED. The prototype includes a PostgreSQL database (876,100 machine-hours), the same data in Google BigQuery, an MQTT edge gateway, four machine-learning models and a live 8-page dashboard.

A. Industrial data volume

Stream	per second	per hour	per day	per month (30 d)	per year
Fleet: 100 machines × 10 sensors × 1 Hz	1,000	3.6 M	86.4 M	2.59 B	**31.54 B**
One 100 Hz vibration sensor (one machine)	100	360,000	8.64 M	259.2 M	**3.15 B**
One 100 Hz sensor on every machine	10,000	36 M	864 M	25.9 B	**315.4 B**

- **The problem with high-frequency data.** A single 100 Hz sensor on every machine produces **10 times** the data of the entire 1 Hz fleet. At the real Bosch rate (2 kHz × 3 axes), raw vibration needs **96 Mbps and 378 TB a year**. That is too much for typical plant internet links and costly to store.
- **Compression alone doesn't solve it.** Lossless compression (Parquet + Zstd), measured on the real data, reached only **5×** on vibration, because vibration is mostly noise.
- **The answer is edge processing:**
- **feature extraction:** 42 features per second (RMS, kurtosis, crest factor, frequency-band energy) give **143× less data**, and the fault signal survives (ROC-AUC **0.98** on real Bosch data);
- **aggregation:** sensor readings are summarised to 1-minute and 1-hour values;
- **noise filtering;**
- **raw snippets only around anomalies.**
- **Result:** fleet bandwidth falls from **19.2 to 0.13 Mbps**.

B. Cloud platform comparison

The comparison was checked against official vendor pages on 21 October 2026, including recent retirements: GCP IoT Core, Azure Time Series Insights, and AWS IoT Analytics and IoT Events.

Criterion	AWS	Microsoft Azure	Google Cloud
Industrial IoT	SiteWise Edge reads OPC UA; several IIoT services retired	**IoT Operations: OPC UA connector, asset discovery**	No first-party IIoT; partner (Litmus) needed
Data ingestion	IoT Core, Kinesis	IoT Hub, Event Hubs, Eventstreams	Pub/Sub
Storage	S3, Timestream	ADLS Gen2, Eventhouse (time-series)	Cloud Storage, BigQuery, Bigtable
Stream processing	Kinesis, Flink (assembled from parts)	**Eventstreams with no-code windows and de-duplication**	Dataflow (code-first)
SQL / structured data	RDS / Aurora PostgreSQL	Azure Database for PostgreSQL	Cloud SQL, BigQuery
Analytics	Athena, QuickSight	KQL, Fabric, **Power BI**	**BigQuery (strongest SQL analytics)**
AI/ML	**SageMaker**	Azure ML, Fabric notebooks	**Vertex AI, BigQuery ML**
Edge computing	**Greengrass, well-documented outage buffering**	IoT Operations on Arc/Kubernetes	Partner edge only
Security	X.509 identities, Mumbai region	OT-grade PKI, Defender for IoT	Data pinned to India regions
Scalability	Very high	Very high	Very high
Integration	SAP via AppFlow	**Microsoft 365, Entra, Power BI, SAP**	Best SAP connector
Ease of use	IIoT now a set of separate parts	Fabric SaaS, common Microsoft skills	BigQuery simple, edge is DIY
Indicative cost (edge design)	₹43k a month	₹80k a month	**₹33k a month**

What it means for the factory: - **Azure** offers the most complete current chain from old controllers to dashboards. - **AWS** has the best-documented edge buffering. - **GCP** has the cheapest and strongest analytics, but no factory-floor layer of its own.

C. Weighted decision matrix (scores 1–5)

Criterion	Weight	AWS	Azure	GCP
Industrial IoT & OT connectivity	15 %	4	5	2
Indicative cost (INR)	13 %	4	2	5
Edge computing & offline operation	12 %	4	4	2
Security, compliance & India residency	12 %	4	4	3
Time-series storage & data lake	10 %	4	5	4
Stream processing & analytics	10 %	3	5	4
AI/ML	10 %	5	4	5
Integration (MES/ERP, Microsoft 365, OPC UA)	8 %	3	5	4
Scalability	5 %	5	5	5
Ease of use & skills	5 %	3	4	3
Weighted total	**100 %**	**3.92**	**4.22**	**3.57**

Recommendation: Microsoft Azure (Central India, Pune), with an edge-first design. The reasons come from the scenario, not only from the score: - **Brownfield controllers:** the plants run old Fanuc and Siemens controllers that speak OPC UA, and Azure reads OPC UA natively. - **Existing tools:** the plants already use Microsoft tools (Power BI, Microsoft 365). - **Location:** the Azure region is in Pune, next to Chakan. - **Weaknesses removed by design:** the edge layer removes Azure's cost and offline-time weaknesses.

How robust the result is: - Azure wins every ±5-point weight shift and **88.7 % of 10,000 random re-weightings**. - AWS wins only if cost carries at least 35 % of the weight. - **Edge batching is the biggest cost lever:** sending one message per reading would cost ₹14–30 lakh a month, against ₹33k–80k a month with edge batching.

D. SQL database design (PostgreSQL 16)

Schema. The main table is `core.machine_data`, with one row per machine per hour: - the required columns: machine_id, ts, temperature, vibration, rpm, current, production_count, machine_status, alarm; - extra columns: reject_count, alarm_code, tool_wear and running_minutes.

It references machine, plant, part and alarm master tables, and CHECK constraints reject impossible values. It holds **876,100 rows**. The same queries were re-run on **Google BigQuery** (Mumbai) and gave identical results.

Question	Answer from the data
Average temperature per machine	36.4–39.1 °C across the 100 machines (hobbing machines run hottest)
Machine with the most parts	**P3-TURN-05**: 199,677 brake discs
Highest average vibration	**P1-VMC-13**: 1.86 mm/s (ISO 10816 zone B, acceptable)
Alarms per machine	4,680 alarm events in total; P1-HMC-22 has the most (75)
Parts on a selected day (16 Jun 2025)	23,974 parts, 381 rejected
Longest operating duration	**P1-HMC-18**: 209 h without stopping
Average RPM while running	VMC 6,888 · HMC 5,334 · grinder 3,168 · turning 2,300 · hobbing 967
Highest rejection rate	**P2-GRIND-06**: 1.97 % (finish-ground gear shafts)

Extra analytics: fleet **OEE 75.3 %** (world-class is 85 %), MTBF/MTTR, alarm Pareto, and vibration anomalies that come before failures.

E. GitHub repository

The repository is at **github.com/TirthankerSingh/industrial-data-platform**. Its structure follows the assignment: `data/`, `python/`, `sql/`, `dashboard/`, `diagrams/`, `README.md` and `report.pdf`. - **README** covers the problem, scenario, datasets, libraries, run steps, SQL setup, key findings and architecture. - **History** has descriptive commits (feat/fix/docs) built up in stages, one part at a time. - **Secrets** (database passwords, cloud keys) are git-ignored and were never committed.

F. Industrial cloud architecture (Azure)

Block	Azure service (prototype component)
CNC / PLC / sensors	Fanuc / Siemens controllers + spindle accelerometer
OPC UA / MQTT	OPC UA connector; MQTT broker with an ISA-95 topic tree (enterprise / plant / area / machine)
Edge gateway	**Azure IoT Operations** on Arc-enabled Kubernetes: 42 features per second, 7-day store-and-forward buffer
Cloud ingestion	IoT Hub / Event Hubs → Fabric Eventstreams
Data lake / time-series	**ADLS Gen2 lakehouse** (bronze, silver, gold) + Fabric Eventhouse
SQL database	Azure Database for PostgreSQL (Flexible Server) (prototype: PostgreSQL 16 and Google BigQuery)
Python analytics / AI	Azure ML / Fabric notebooks (prototype: Isolation Forest, Random Forest)
Dashboard	Power BI (prototype: Streamlit, live)
Manufacturing decision	Maintenance before failure, tool changes based on condition, OEE improvement

G. Edge versus cloud

At the edge	In the cloud
2 kHz vibration processing into features	Historical and fleet-level analysis
Immediate alarms and local machine protection	Long-term storage (lakehouse)
Noise filtering and data aggregation	AI/ML model training and management
Store-and-forward buffering during outages	Management dashboards and remote access

Engineering justification: - **Latency:** machine protection needs millisecond responses, which the cloud cannot guarantee. - **Reliability:** a plant must keep running when its internet link fails. Our gateway test lost **0 of 482 messages** through two outages. - **Bandwidth:** sending features instead of raw signals cuts the data 143×. - **Cost:** the cloud bill stays at ₹33k–80k a month instead of lakhs. - **Cybersecurity:** the OT network makes only outbound, encrypted connections, following IEC 62443 zones. - **Scalability:** adding a plant means adding another identical edge node.

The cloud is where scale pays off: the full fleet history, model training and dashboards for the whole company.

Storage pattern and results

Storage pattern: a lakehouse, chosen over a pure lake or a pure warehouse. - Over 99 % of the bytes are raw vibration, which belongs in a cheap lake. - Clean, de-duplicated tables serve dashboards and AI from the same copy of the data.

Results: - **Bosch anomaly detection:** ROC-AUC **0.98**, holding when trained on 2019 data and tested on 2021. - **24-hour failure warning:** works on the benchmark data. It scores almost perfectly only because that data is simulated. - **Estimated business value:** about **₹5.8 crore a year**. - **Live dashboard:** reads Google BigQuery in Mumbai through a read-only service account. If BigQuery is unavailable, it falls back to a Parquet snapshot.